# Plant Safety Procedures

EHS-SOP-001 | Revision D | Effective 2026-03-01

| Field | Value |
|---|---|
| Facility | Northfield Components Inc. - Plant 2 (Riverside) |
| Document owner | Environment, Health and Safety (EHS) Manager |
| Approved by | Plant Manager; EHS Manager |
| Applies to | All employees, contractors and visitors on the Plant 2 production floor |
| Review cycle | Annually, and after any serious incident or process change |

> **Note:** FICTIONAL SAMPLE DOCUMENT - for software development and testing only. Names, equipment, part numbers and values are invented.

## 1. Purpose and Scope

This procedure sets the minimum safety requirements for work on the Plant 2 production floor: Line 1 (CNC machining), Line 2 (pressing and welding), Line 3 (assembly and packing) and the support areas (compressor room, parts-washing bay and shipping dock). Where an equipment manual or quality standard conflicts with this procedure, **this procedure takes precedence**.

Related documents: MM-100 Equipment Maintenance Manual; QC-STD-200 Quality Control Standards; Safety Data Sheets (SDS) for all chemicals on site.

## 2. Roles and Responsibilities

| Role | Responsibilities |
|---|---|
| Floor supervisor | Enforces this procedure on their shift; verifies PPE and lockout compliance; leads the pre-shift safety talk (5 minutes); completes incident form EHS-F-07; performs evacuation headcount. |
| Employee / operator | Follows this procedure; inspects own PPE before use; stops work and reports any unsafe condition; never bypasses a guard or interlock. |
| Maintenance technician | Applies lockout/tagout before any servicing; removes and reinstalls guards; returns equipment to service only after guards and interlocks are verified. |
| EHS department | Owns this procedure; trains employees; investigates incidents; manages permits, SDS library and PPE stock. |
| Contractors | Follow this procedure plus any contract-specific safety plan; must be escorted until site orientation is completed. |

Every employee has **stop-work authority**: anyone may stop a task they believe is unsafe, without penalty. The supervisor must review the concern before work restarts.

# 3. General Shop-Floor Rules

- Walk only in the green-marked pedestrian walkways. Yellow lanes are reserved for forklifts.
- No personal phones or earbuds on the production floor outside the break rooms.
- Loose clothing, jewelry (including rings) and hair longer than shoulder length must be secured near rotating equipment.
- Keep aisles, electrical panels (0.9 m / 36 in clearance), fire extinguishers and eyewash stations clear at all times.
- Report all injuries, near misses and unsafe conditions to the floor supervisor immediately (see Section 11).
- No food or drink at workstations. Hydration stations are located at the end of each line.

# 4. Personal Protective Equipment (PPE)

## 4.1 Baseline PPE

The following PPE is required for everyone entering the production floor, including visitors:

- Safety glasses with side shields meeting ANSI Z87.1.
- Safety-toe footwear meeting ASTM F2413.
- Hearing protection (plugs or muffs) in all of Line 2 and the compressor room, where noise exceeds 85 dBA. Hearing protection is optional elsewhere unless posted.

## 4.2 Task-Specific PPE Matrix

The PPE below is required **in addition to** baseline PPE.

| Task / area | Additional PPE required | Notes |
|---|---|---|
| CNC lathe operation (CNC-201/202/203) | None beyond baseline. NO gloves while the spindle is turning. | Gloves can be caught by rotating parts. |
| Removing chips and swarf | Cut-resistant gloves, ANSI A4 or higher | Spindle must be stopped. Use a chip hook, never bare hands. |
| Handling machined or stamped parts with sharp edges (A-102, B-220, C-310) | Cut-resistant gloves, ANSI A2 or higher | Applies to inspection and packing too. |
| Hydraulic press HP-400 operation | Hearing protection (Line 2 area) | Two-hand controls and light curtain must be active. |
| Die change on HP-400 | Cut-resistant gloves A4, safety-toe boots with metatarsal guard | Lockout required (Section 5). |
| Robotic welding cell RW-510 (entry for tip change or teach) | Welding jacket (FR), leather gloves, face shield over safety glasses | Robot must be in locked-out or teach mode at reduced speed (max 250 mm/s). |
| Manual welding / grinding | Auto-darkening helmet shade 10-13, FR jacket, leather gauntlet gloves | Hot work permit required outside the welding bay (Section 8.1). |
| Solvent parts washer PW-1 (normal use) | Nitrile gloves (minimum 15 mil), chemical splash goggles, chemical-resistant apron | Local exhaust ventilation must be running. |


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| Task / area | Additional PPE required | Notes |
|---|---|---|
| PW-1 solvent change or tank cleaning, or exhaust ventilation not running | All of the above plus half-face respirator with organic vapor cartridges | Respirator users must be fit-tested within the last 12 months. |
| Coolant mixing / sump cleaning (Line 1) | Nitrile gloves, splash goggles | Coolant sump pit entry is confined space (Section 8.2). |
| Compressed air for cleaning | Face shield over safety glasses | Nozzle must be OSHA-compliant, under 30 psi dead-end pressure. Never aim at a person. |
| Forklift operation | High-visibility vest | Seat belt must be worn. |

**WARNING:** Never wear gloves when operating a lathe, drill or any machine with an exposed rotating spindle. Gloves are allowed only when the spindle is fully stopped.

### 4.3 PPE Inspection and Replacement

Inspect PPE before each use. Replace cut-resistant gloves when torn or saturated with oil, safety glasses when scratched enough to impair vision, and respirator cartridges every 40 hours of use or immediately if the wearer smells solvent. Replacement PPE is issued at the tool crib on any shift.

## 5. Lockout/Tagout (LOTO) - Control of Hazardous Energy

### 5.1 When Lockout Is Required

Lockout is required before any person places any part of their body into a point of operation or danger zone, and for:

- Servicing, maintenance, lubrication or repair (unless the MM-100 task is explicitly rated as "running lubrication" from outside the guard).
- Clearing jams on conveyor CV-3, HP-400, the CNC lathes or the case packer.
- Removing, adjusting or replacing any machine guard.
- Die changes, tooling changes that require reaching into the machine, and welding tip changes on RW-510.
- Any work on electrical, hydraulic or pneumatic components.

**NOTE:** There is **no minor-servicing exception** for hydraulic press HP-400. Every entry into the HP-400 die area requires full lockout and ram blocks.


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## 5.2 The Six-Step Lockout Procedure

1. **Prepare and notify.** Identify all energy sources using the table in Section 5.3. Tell the operator and the floor supervisor that the machine is being locked out.
2. **Shut down.** Stop the machine using its normal stop control.
3. **Isolate.** Open the main disconnect and close or bleed each additional energy-isolating device (air, hydraulic, gravity).
4. **Apply locks and tags.** Each worker applies their own personal red lock and a tag showing their name, date and reason. Never use another person's lock.
5. **Release stored energy.** Bleed accumulators and air lines, block or lower raised parts, and let moving parts come to a complete stop.
6. **Verify zero energy.** Press the start button and check gauges to confirm the machine cannot start. Return controls to OFF afterward.

## 5.3 Energy Sources by Equipment

| Equipment | Energy sources | Isolation points |
|---|---|---|
| Hydraulic press HP-400 (Line 2) | Electrical 480 V; hydraulic (accumulator up to 3,000 psi); gravity (ram) | Disconnect DS-400; accumulator bleed valve HV-2 (verify gauge reads 0 psi); two ram safety blocks from the red storage rack |
| Conveyor CV-3 (Line 3) | Electrical 480 V; gravity on incline section | Disconnect DS-3C; chock the incline belt with the yellow belt clamp |
| CNC lathes CNC-201/202/203 (Line 1) | Electrical 480 V; pneumatic 90 psi (chuck) | Machine main disconnect; air valve AV-2xx with lockable dump |
| Robotic welder RW-510 (Line 2) | Electrical 480 V; pneumatic; weld power; robot arm gravity | Disconnect DS-510; air valve AV-510; place arm in home position before lockout |
| Air compressor AC-1 (compressor room) | Electrical 480 V; stored air in 500 gal receiver | Disconnect DS-AC1; close outlet valve and drain receiver to 0 psi |
| Parts washer PW-1 | Electrical 120 V (pump) | Unplug and apply plug lockout cover |
| Case packer CP-3 (Line 3) | Electrical 480 V; pneumatic 80 psi | Disconnect DS-CP3; air valve AV-CP3 |

## 5.4 Group Lockout and Shift Change

When several people work on the same equipment, use the group lockbox at the line station: the lead technician locks the isolation points and places the keys in the lockbox, then every worker applies a personal lock to the lockbox. At shift change, the **oncoming worker applies their lock before the off-going worker removes theirs**, so the equipment is never left without protection.

## 5.5 Removing Another Person's Lock

A lock may be removed only by the person who applied it. If that person has left the site, the lock may be removed only with written authorization from both the Plant Manager and the EHS Manager, after documented attempts to contact the worker, and the worker must be told before they resume work.


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# 6. Machine Guarding

- Never operate a machine with a guard removed, damaged, or bypassed, or with an interlock defeated.
- Guards may be removed, adjusted or replaced only by maintenance technicians, and only with the machine locked out (Section 5).
- **A guard must never be removed or replaced while the line is running.** Only the machine being worked on must be locked out; adjacent machines on the same line may keep running only if they do not share an energy source and a barricade is placed at least 1 m (3 ft) from the work area.
- After reinstalling a guard, the technician tests the interlock and the supervisor signs the return-to-service line on the work order before production restarts.
- If a guard is found damaged or missing, stop the machine, hang a yellow "DO NOT OPERATE" tag and notify maintenance and the supervisor.

## 6.1 Start-of-Shift Safeguard Checks

| Equipment | Check | Pass criterion |
|---|---|---|
| HP-400 light curtain | Pass the 30 mm test rod slowly through the full height of the curtain while the press is in inch mode | Press stops every time the rod breaks the beam |
| HP-400 two-hand control | Release one button during the stroke | Ram stops immediately |
| CNC lathe door interlocks | Open the door with spindle commanded at low speed | Spindle will not start or stops within 2 s |
| RW-510 cell gate | Open the gate in auto mode | Robot stops; fault shown on pendant |
| CV-3 emergency pull-cords | Pull cord at each of 3 stations (one station per day, rotating) | Conveyor stops; reset required |

Any failed check means the machine must not run. Tag it out and notify maintenance; record the failure on the shift log.

# 7. Hazardous Materials and Chemicals

## 7.1 Safety Data Sheets and Labels

SDS for every chemical are kept in the yellow SDS binder at each line station and on the plant intranet. All secondary containers (spray bottles, transfer cans) must carry a label with the product name and hazard pictograms.

## 7.2 Parts Washer PW-1 Solvent

PW-1 uses SafeClean 150, a petroleum-naphtha solvent with a flash point of 61 °C (142 °F).

- No open flames, grinding, welding or other ignition sources within 6 m (20 ft) of PW-1.
- Keep the lid closed when not actively cleaning parts. The lid has a fusible link that closes it automatically in a fire.
- Local exhaust ventilation must be running during use; if it is not, stop use or wear a respirator (see Section 4.2).


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- Store spare solvent only in the flammable-liquids cabinet in the parts-washing bay (maximum 60 gal / 227 L in the cabinet).
- Used solvent goes in the red drum labeled "Hazardous Waste - Spent Solvent". Never pour solvent into floor drains.

### 7.3 Spill Response

| Spill size | Who responds | Action |
|---|---|---|
| Small: under 5 L (about 1 gal) and not near a drain | Trained employee in the area | Wear task PPE; contain with the spill kit at the line station; place used absorbent in the hazardous waste drum; tell the supervisor. |
| Large: 5 L or more, any amount entering a floor drain, or any unknown chemical | EHS emergency response | Leave the area, warn others, call emergency extension 5555; the supervisor keeps people out until EHS clears the area. |

## 8. Work Permits

### 8.1 Hot Work Permit

Welding, cutting, brazing or grinding outside the designated welding bay requires a hot work permit from EHS. The permit is valid for one shift only. Remove combustibles within 11 m (35 ft) or cover them with fire blankets. A fire watch with an extinguisher must stay during the work and for **30 minutes** after it ends.

### 8.2 Confined Space Entry

The Line 1 coolant sump pit and the inside of the 500 gal compressor receiver are permit-required confined spaces. Entry requires an EHS entry permit, an attendant outside the space at all times, and atmospheric testing before and during entry. Acceptable atmosphere: oxygen 19.5% to 23.5%, flammable gas below 10% of LEL, hydrogen sulfide below 10 ppm and carbon monoxide below 35 ppm.

## 9. Emergency Response

### 9.1 Emergency Contacts

| Situation | Contact |
|---|---|
| Fire, serious injury, large spill, gas leak | Emergency extension 5555 (staffed 24/7); then 911 if directed |
| First aid needed | Floor supervisor; first-aid kits at each line station and in the break room |
| EHS on-call (nights and weekends) | Via the security desk, extension 5000 |

### 9.2 Fire

Use the RACE response: **R**escue anyone in immediate danger, **A**larm (pull station and call 5555), **C**onfine the fire by closing doors, **E**xtinguish only if the fire is small and you are trained, or Evacuate. Operate extinguishers using PASS: Pull, Aim, Squeeze, Sweep. Never fight a fire involving PW-1 solvent larger than a wastebasket.


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### 9.3 Evacuation

When the alarm sounds, stop machines with the normal stop control if it is safe to do so and leave by the nearest exit. The assembly point is **Area B in the north parking lot**. The floor supervisor takes the shift roster, completes a headcount within 10 minutes and reports missing people to the incident commander. Nobody re-enters until EHS gives the all-clear.

### 9.4 Injuries and Chemical Exposure

- Chemical in the eyes: flush at the nearest eyewash station for at least 15 minutes, holding the eyelids open, then seek medical evaluation.
- Chemical on the skin: remove contaminated clothing and rinse for 15 minutes; use the safety shower for large exposures.
- Cuts and lacerations: apply direct pressure; anything needing more than first aid goes to the clinic per the supervisor.
- Eyewash stations and safety showers are activation-tested weekly by the area supervisor and tagged with the test date.

## 10. Forklift and Pedestrian Safety

- Only certified operators may drive forklifts. Certification is renewed every 3 years, or after an incident or near miss.
- Complete the pre-shift forklift checklist (brakes, horn, lights, forks, chains, hydraulic leaks). A failed item means the truck is tagged out.
- Maximum speed on the production floor is 5 mph (8 km/h); sound the horn at intersections and blind corners.
- Pedestrians must make eye contact with the driver before crossing a forklift lane.
- Never stand or walk under raised forks, and never ride on forks or pallets.

## 11. Incident and Near-Miss Reporting

1. Report every injury, near miss, property damage or spill to the floor supervisor immediately, before the end of the shift.
2. The supervisor secures the scene and completes incident form **EHS-F-07** within 24 hours.
3. Serious injuries (hospital treatment, amputation, loss of an eye, or loss of consciousness) must be reported by the supervisor to the EHS Manager **within 1 hour**.
4. EHS leads a root-cause investigation within 5 working days and assigns corrective actions.


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## 12. Training Requirements

| Training | Who | Frequency |
|---|---|---|
| Safety orientation | All new employees and contractors | Before starting work |
| Lockout/tagout (authorized) | Maintenance technicians, setup technicians | Initial and annually |
| Lockout/tagout (affected) | Operators and supervisors | Initial and every 2 years |
| Hazard communication / SDS | All employees | Initial and annually |
| Respirator fit test | PW-1 users and anyone using respirators | Annually |
| Forklift operator | Forklift operators | Every 3 years |
| First aid / CPR | At least 2 per shift per line | Every 2 years |
| Confined space entry | Entrants, attendants, supervisors | Initial and annually |


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